

EMILY LIANG

Personal website: emilyliangyr.com

☎ 682-205-4481

✉ emilyliangyr@gmail.com

🌐 [linkedin.com/in/emilyliangyr](https://www.linkedin.com/in/emilyliangyr)

🐙 github.com/Tacocat0254

Education

The University of Texas at Austin

May 2028 (Expected)

B.S. in Computer Science and Double major in Mathematics + Minor in Business

Austin, TX

Honors: Turing Scholars Honors Program GPA: 3.87

Relevant Coursework

- Programming Languages (H)
- Operating Systems (H)
- Computer Vision
- Differential Equations
- Artificial Intelligence (H)
- Computer Architecture (H)
- Software Engineering
- Linear Algebra
- Algorithms (H)
- Discrete Math (H)
- Stochastic Processes
- Probability
- Systems of ML (H)
- Data Structures (H)
- Number Theory

Experience

Quantitative Developer Intern

May. 2026 – Present

Base Power Company

Austin, TX

- Implemented an optimization-based battery distribution algorithm that improved expected profit by approximately \$4 per kWh under real-world electrical market (ERCOT) constraints.
- Built simulation and xval algorithms to evaluate battery dispatch strategies under varying ERCOT market conditions.
- Designed and executed statistical experiments to quantify strategy performance and guide optimization decisions.

Longhorn Racing Internal Combustion

Aug. 2024 – Present

Simulation and Validation — System Lead

Austin, TX

- Developed a vehicle dynamics and lap-time simulation framework to model system-level performance and evaluate 10+ engineering design tradeoffs.
- Led a team of 7 engineers to development of simulation, telemetry, and validation infrastructure used to analyze thousands of laps of vehicle data and calibrate predictive performance models.

Software Engineering Intern

Sept. 2023 – May 2024

Lockheed Martin Aeronautics

Fort Worth, TX

- Collaborated with the Air Force Research Laboratory, L3 Harris, and Raytheon on research efforts for star-link on F35s.

Projects

JPEB: 16-Bit Computer | *Rust, Haskell, C, Verilog, Python, Assembly*

April 2025

- Collaborated in a team of four to design and implement a custom 16-bit computer from scratch, including a custom ISA inspired by Dr. Bruce Jacob's RiSC-16.
- Developed a 6-stage pipeline processor, deployed it on an FPGA, including hazard detection, data forwarding, and pipeline stalling and flushing. Extended an existing compiler and assembler to support game development.
- Implemented the Chrome Dinosaur and Snake games using the custom 16-bit ISA; integrated graphics rendering into the emulator, increasing test coverage and validation efficiency.

Network Protocol Stack Implementation | *Go*

Nov. 2025

- Implemented a reliable transport protocol featuring ACK-based retransmission, packet sequencing, sliding-window flow control, and timeout recovery mechanisms over an unreliable network channel.
- Developed routing and forwarding logic for packet transmission, enabling reliable end-to-end communication under packet loss and network contention.

x86 Multi-Core Operating System Kernel | *C++*

Aug. 2025 - Dec. 2025

- Built a preemptive multicore x86 operating system kernel supporting concurrent execution and synchronization primitives.
- Implemented virtual memory with demand paging, page caching, swapping, and memory-mapped files.
- Developed kernel-level process, thread, file-system, and memory-management system calls.

Simulation and Telemetry Server | *Python*

Aug. 2025 - Present

- Implemented full-stack website for simulation and drive day data dashboard display for design system use.
- Designed validation plans for drive days to calibrate model parameters from empirical data to match expected behavior.
- Collaborate with a team of five to integrate Laptime Simulation, Kinematics Model, Tire model, and data display.

Technical Skills / Additional Information

Languages: C/C++, Python, Go, Java, HTML/CSS, JavaScript/TypeScript, Verilog, x86, ARM Assembly, LaTeX

Developer Tools: Git, Jupyter Notebook, Google Colab, VS Code, Docker, PyCharm, IntelliJ, Android Studio

Libraries: pandas, NumPy, Matplotlib, Scikit-Learn, TensorFlow, Keras